Unified Mentor Internship Project Report

E-Commerce Furniture Sales Analysis (2024) — Excel Dashboard (Primary) + Basic ML (Summary)

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Company: Unified Mentor Pvt. Ltd.
Project Duration: (Add your dates)
Tools Used: Microsoft Excel 365, Google Colab (Python), pandas, scikit-learn (basic)
Dataset: Company-provided CSV — *E-commerce furniture sales data (2024)*

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1. Executive Summary

This project focuses on analyzing an e-commerce furniture sales dataset for 2024 and delivering a professional, interactive Excel dashboard for business reporting. The dataset was cleaned, duplicates were removed, and meaningful analysis-ready columns were prepared. Using Excel 365, PivotTables, PivotCharts, slicers, and KPI cards were built to create a complete dashboard that updates dynamically based on selected filters.

In addition, a **basic Machine Learning (ML) model** was created in Google Colab as a secondary deliverable to demonstrate how Python can be used to predict "units sold" from pricing and discount-related features. The ML section is kept beginner-friendly and concise, while the main focus of the report is Excel analytics and dashboarding.

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2. Project Objectives

2.1 Excel Analytics Objectives (Main)

- Clean and prepare the dataset for analysis in Excel.
- Create an interactive dashboard showing:
  - Total units sold
  - Average price
  - Average discount
  - Category-wise performance (Furniture types)
  - Shipping type effect (Free vs Paid)
  - Top selling products
- Build slicers so that **one click updates all charts and KPIs**.

2.2 Python/ML Objective (Secondary)

- Build a basic regression model to predict "sold" using simple numeric features (price, original price, discount).
- Compare two beginner-level models and understand performance.

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3. Dataset Overview

The dataset contains product-level information for furniture listings on an e-commerce platform. Each row represents one product listing with attributes such as product title, current price, original price, discount percentage, sold units, shipping type, and shipping fee.

Key Columns Used for Excel Reporting

- **productTitle** – Product name/title
- **price** – Current price
- **original_price_fill(in dollars)** / **originalPrice** – Original price (used for discount logic)
- **discount_percentage** – Discount value (stored as number like 47.7)
- **sold** – Units sold
- **shipping_type** – Free / Paid
- **shipping_fee** – Mostly 0 in this dataset
- **Furniture_Type** (New Column) – Extracted category based on productTitle (Sofa, Chair, Desk, etc.)

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4. Data Cleaning & Preparation in Excel

4.1 Duplicate Removal

- Duplicate records can create wrong results in PivotTables (double counting).
- Duplicates were removed using Excel's **Remove Duplicates** feature.
- This ensured that analysis and dashboard values are accurate.

4.2 Converting Data into Excel Table

- The dataset was converted into an Excel Table for:
  - Auto-expanding ranges
  - Easy PivotTable creation
  - Reliable slicer connections

- A table format also ensures clean references and structured analysis.

4.3 Handling Original Price Column

- The raw original price column had missing values and was not consistent.
- Instead of removing it, it was:
  - Hidden (to avoid confusion)
  - A cleaned/fill version was used for analysis where needed
- This kept the dataset usable without losing information.

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5. Feature Engineering in Excel (Furniture_Type Column)

5.1 Why Furniture_Type was needed

The dataset does not directly provide a ready "category" column. But category-wise analysis is essential for dashboards. Therefore, a new column **Furniture_Type** was created by extracting the furniture category from **productTitle**.

5.2 Approach Used

A keyword mapping method was applied:

- A separate **Helper** sheet was created with two columns:
  - **Keyword**
  - **FurnitureType**
- Example keywords: "sofa", "chair", "desk", "coffee table", "bed frame", etc.
- A formula (XLOOKUP + SEARCH logic) automatically matches the product title with keywords and assigns the correct category.

5.3 Benefit

This allowed:

- Category-wise PivotTables
- Category-wise slicers
- A more impressive and real-world dashboard segmentation

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6. PivotTables and Charts Created (Excel Deliverables)

The analysis was built using multiple PivotTables (one per chart) to keep the dashboard clean and professional. Each PivotTable supports a PivotChart and is connected to the same slicer.

Chart 1: Units Sold by Furniture Type (Bar Chart)

Purpose: Identify which furniture category sells the most.
Pivot Logic:

- Rows: Furniture_Type
- Values: Sum of sold
  Outcome: Shows the best-selling categories instantly.

Chart 2: Average Price by Furniture Type (Column Chart)

Purpose: Compare average pricing across categories.
Pivot Logic:

- Rows: Furniture_Type
- Values: Average of price
  Outcome: Highlights premium vs budget categories.

Chart 3: Average Discount (%) by Furniture Type (Column Chart)

Purpose: Compare discount strategies for each category.
Pivot Logic:

- Rows: Furniture_Type
- Values: Average of discount_percentage
  Note: Discount values were formatted correctly as numbers (not wrong percentage conversion).

Chart 4: Units Sold by Shipping Type (Pie / Donut)

Purpose: Understand effect of Free vs Paid shipping on sales volume.
Pivot Logic:

- Rows: shipping_type
- Values: Sum of sold
  Outcome: Free shipping was the majority, making this insight valuable.

Chart 5: Top 10 Products by Units Sold (Bar Chart)

Purpose: Identify the highest-performing products.
Pivot Logic:

- Rows: productTitle
- Values: Sum of sold
- Filter: Top 10 by Sum of sold
  Outcome: Shows exact products driving maximum sales.

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7. Dashboard Development (Main Deliverable)

7.1 Dashboard Components

The final dashboard was created on a dedicated sheet **Dashboard** and includes:

A) KPI Cards (Top Row)

1. **Total Units Sold**
2. **Average Discount (%)**
3. **Average Price**

These KPIs update dynamically using slicer connections.

B) One Common Slicer (Professional Feature)

- **Furniture_Type slicer** was created and connected to all PivotTables.
- This means selecting “Sofa” or “Chair” updates:
    - All charts
    - KPI cards
    - Insights values

This is a key professional dashboard feature.

C) Charts Section

All 5 charts were placed in a clean layout under the KPIs.

D) Insights Box

A dedicated Insights panel summarises key findings such as:

- Best selling furniture type
- Highest average price type
- Highest discount type
- Shipping observation
- Top product by units sold

This helps evaluators quickly understand results without exploring charts deeply.

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8. Key Insights from Analysis (Example from Dashboard)

(These insights update based on slicer selection.)

- **Best-selling furniture type:** Chair

- **Highest average price type:** Patio Furniture Set
- **Highest discount type:** Desk
- **Shipping observation:** Free shipping dominates sales
- **Top product by units sold:** Displayed via Top 10 product chart

(These insights demonstrate category-level performance, pricing strategy, and shipping impact.)

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9. Basic Machine Learning Summary (Secondary Deliverable)

(Kept short as requested)

To demonstrate predictive analytics, a simple machine learning workflow was built in **Google Colab** using Python.

9.1 What was done in Colab

- Loaded CSV using pandas
- Cleaned numeric columns (price, originalPrice)
- Filled missing values using simple beginner logic
- Created one engineered feature:
  - **discount_percent**
- Performed basic EDA plots:
  - Sold distribution
  - Price distribution
  - Price vs sold

9.2 Models Tried (Simple)

Two regression models were trained:

- **Linear Regression**
- **Random Forest Regressor**

To improve stability:

- Outliers were capped (winsorization at 99th percentile)
- Target was log-transformed to handle skewness

9.3 Output

The ML part helped understand how pricing and discount affect predicted sold units. This was a basic extension to show additional capability beyond Excel.

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10. Challenges Faced & Solutions

Challenge 1: CSV saving removed all dashboard sheets

- CSV cannot store PivotTables, charts, slicers, or multiple sheets.
- **Solution:** File was saved correctly as **.xlsx** Excel workbook.

Challenge 2: Extracting furniture type from product titles

- Thousands of rows, manual categorization impossible.
- **Solution:** Helper sheet + keyword mapping + formula-based automation.

Challenge 3: Shipping fee mostly zero

- Shipping fee chart would be dull.
- **Solution:** Used **shipping_type** (Free vs Paid) as a clearer business insight.

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11. Conclusion

This project successfully delivered a complete Excel-based analytics solution for e-commerce furniture sales data. The final output is an interactive, professional dashboard with KPI cards, slicers, and well-structured charts. The dashboard allows stakeholders to explore category-wise and product-wise performance easily and understand key factors like pricing, discounts, and shipping type impact.

Additionally, a basic ML workflow was performed in Google Colab to demonstrate predictive analytics skills using simple regression models. Overall, the project meets internship expectations and showcases strong Excel analytics and reporting capability.

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12. Deliverables Submitted

Link of excel sheet: ecommerce_furniture_dataset_2024 (1).xlsx

1. **Excel Workbook (.xlsx)** containing:
    - Clean dataset table
    - Helper sheet for categories
    - PivotTables + PivotCharts
    - Dashboard with slicer + KPIs + insights
2. **Google Colab Notebook** containing:

- Data cleaning
- EDA plots
- Basic regression models (LR + RF)

Link of Google Collab:
https://colab.research.google.com/drive/1Nmk76wK_C9UK2f0vMgj5mVTVh17NOEAr?usp=sharing